

CS 229, Winter 2026

Problem Set #3

Due Wednesday, February 18 at 11:59 pm on Gradescope.

Notes: (1) These questions require thought, but do not require long answers. Please be as concise as possible.

(2) If you have a question about this homework, we encourage you to post your question on our Ed forum, at <https://edstem.org/us/courses/90830/discussion/>.

(3) If you missed the first lecture or are unfamiliar with the collaboration or honor code policy, please read the policy on the course website before starting work.

(4) The due date is Wednesday, February 18 at 11:59 pm. If you submit after Wednesday, February 18 at 11:59 pm, you will begin consuming your late days. The late day policy can be found in the course website: **Course Logistics and FAQ**.

All students must submit an electronic PDF version of the written solutions. We highly recommend typesetting your solutions via L^AT_EX.

Honor code: We strongly encourage students to form study groups. Students may discuss and work on homework problems in groups. However, each student must write down the solution independently, and without referring to written notes from the joint session. Each student must understand the solution well enough in order to reconstruct it by him/herself. It is an honor code violation to copy, refer to, or look at written or code solutions from a previous year, including but not limited to: official solutions from a previous year, solutions posted online, and solutions you or someone else may have written up in a previous year. Furthermore, it is an honor code violation to post your assignment solutions online, such as on a public git repo. We run plagiarism-detection software on your code against past solutions as well as student submissions from previous years. Please take the time to familiarize yourself with the Stanford Honor Code¹ and the Stanford Honor Code² as it pertains to CS courses.

¹<https://communitystandards.stanford.edu/policies-and-guidance/honor-code>

²<https://web.stanford.edu/class/archive/cs/cs106b/cs106b.1164/handouts/honor-code.pdf>

1. [10 points] **Convolution and its Properties as a Structured Linear Map**

Let $n, m \in \mathbb{N}$ with $1 \leq m \leq n$. Let $k = (k_1, \dots, k_m) \in \mathbb{R}^m$ be a **nonzero** 1D filter, and define the valid (no padding) convolution map $T_k : \mathbb{R}^n \rightarrow \mathbb{R}^{n-m+1}$ by

$$(T_k(x))_t = \sum_{i=1}^m k_i x_{t+i-1}, \quad t = 1, \dots, n - m + 1.$$

Let $A_k \in \mathbb{R}^{(n-m+1) \times n}$ denote the matrix such that $T_k(x) = A_k x$.

- (a) [4 points] **Structure.** Write A_k explicitly (entrywise). Prove that A_k is Toeplitz (constant along diagonals). An example of a Toeplitz matrix is as follows:

$$\begin{bmatrix} a & b & 0 & 0 \\ c & a & b & 0 \\ 0 & c & a & b \end{bmatrix}.$$

- (b) [6 points] **Rank.** Prove that if $k \neq 0$, then A_k has **full row rank**:

$$\text{rank}(A_k) = n - m + 1.$$

Hint: Pick the *first nonzero* filter coefficient k_j and argue about a leftmost pivot position in each row.

2. [18 points] Decision Trees and Gini Loss

When growing a decision tree, we split the input space in a greedy, top-down, recursive manner. Given a parent region R_{par} , we can choose a split $s_{\text{par}}(j, t)$ which yields two child regions $R_l = \{X \mid x_j < t, X \in R_{\text{par}}\}$ and $R_r = \{X \mid x_j \geq t, X \in R_{\text{par}}\}$. Assuming we have defined a per region loss $L(R)$, at each branch we select the split that minimizes the weighted loss of the children:

$$\min_{j,t} \frac{|R_l|L(R_l) + |R_r|L(R_r)}{|R_l| + |R_r|}$$

For the problems below, assume we are dealing with **binary classification** and that there are no degenerate cases where positive and negative datapoints overlap in the feature space.

$$p_m := \text{the proportion of positive examples in region } R_m, \quad p_m \in [0, 1]$$

When performing classification, a commonly used loss is the Gini loss:

$$G(R_m) = G(p_m) = 2p_m(1 - p_m)$$

However, we are oftentimes more interested in optimizing the final misclassification loss:

$$M(R_m) = M(p_m) = 1 - \max\{p_m, 1 - p_m\} \quad (1)$$

- (a) [5 points] Show that for any given split, the weighted Gini loss of the children cannot exceed that of the parent. (**Hint:** first show that the Gini loss is strictly concave. And then use the fact that G is strictly concave meaning:

$$\forall p_l \neq p_r, \forall t \in (0, 1) : \quad G(tp_l + (1 - t)p_r) > tG(p_l) + (1 - t)G(p_r)$$

- (b) [5 points]

- (i) List out the conditions on the cardinality and proportion of positive and negative samples on the children where Gini loss will stay the same after a split.
- (ii) Show why these do not violate the strict concavity of the Gini loss.
- (iii) Briefly explain why these cases do not prevent a fully grown tree from achieving zero Gini loss.

(**Hint:** Recall the definition of strict concavity).

- (c) [4 points] If instead we use misclassification loss, what additional case causes the loss to stay the same after a split? Show why this is (hint: you may find it useful to define $N_m = |R_m|$ and N_m^+ as the number of positive examples present in R_m).
- (d) [4 points] Consider a training set X . In bootstrap sampling, each time we draw a random sample Z of size N from the training data and obtain $Z_1, Z_2, \dots, Z_B$ after B times, i.e. we generate B different bootstrapped training data sets. If we apply bagging to regression trees, each time a tree $T_i (i = 1, 2, \dots, B)$ is grown based on the bootstrapped data Z_i , and we average all the predictions to get:

$$\hat{T}(x) = \frac{1}{B} \sum_{i=1}^B T_i(x)$$

Now, if $T_1, T_2, \dots, T_B$ are independent of each other, but each has the same variance σ^2 , the variance of the average $\hat{T}$ is σ^2/B . However, in practice, the bagged trees could be similar to each other, resulting in correlated predictions. Assume $T_1, T_2, \dots, T_B$ still share the same variance σ^2 , but have a positive pairwise correlation ρ . We define the correlation between two random variables as:

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}}$$

Thus, we have $\rho = \text{Corr}(T_i(x), T_j(x)), i \neq j$.

Show that in this case, the variance of the average is given by:

$$\text{Var}\left(\frac{1}{B} \sum_{i=1}^B T_i(x)\right) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$$

3. [12 points] AdaBoost

Consider building an ensemble of decision stumps f_t with the AdaBoost algorithm,

$$F(x) = \text{sign} \left(\sum_{t=1}^T \hat{w}_t f_t(x) \right).$$

Figure 1 displays a 2-dimensional training dataset, as well as the first stump chosen. A stump predicts binary $+1/-1$ values, and depends only on one coordinate value (the split point). The little arrow indicates the positive side where the stump predicts $+1$. All points start with uniform weights.

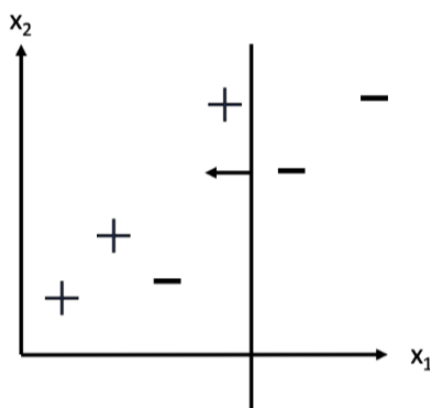


Figure 1: 2-dimensional labeled data, where '+' corresponds to class $y = +1$ and '-' corresponds to class $y = -1$. The decision boundary for the first decision stump is shown. The arrow points in the positive direction from this decision boundary.

- [4 points] Circle all the point(s) in Figure 1 whose weight(s) will increase as a result of incorporating the first stump (the weight update due to the first stump). Briefly explain your selection(s).
- [4 points] Draw a possible stump that we could select at the next boosting iteration. You need to draw both the decision boundary and its positive orientation. The answer may not be unique and any plausible solution is acceptable. No justification is required.
- [4 points] Will the second stump receive higher coefficient in the ensemble than the first? In other words, will $\hat{w}_2 > \hat{w}_1$? Briefly explain your answer. (No calculation should be necessary.)

4. [20 points] AdaBoost Performance

We learned about boosting in lecture. Statistician Kevin Murphy claims that “It can be shown that, as long as each base learner has an accuracy that is better than chance (even on the weighted dataset), then the final ensemble of classifiers will have higher accuracy than any given component.” We will now verify this in the AdaBoost framework.

- (a) [3 points] Given a set of n observations $(x^{(i)}, y^{(i)})$ where $y^{(i)}$ is the label $y^{(i)} \in \{-1, 1\}$, let $f_t(x)$ be the weak classifier at step t and let $\hat{w}_t$ be its weight. First we note that the final classifier after T steps is defined as

$$F(x) = \text{sign} \left\{ \sum_{t=1}^T \hat{w}_t f_t(x) \right\} = \text{sign}\{f(x)\},$$

where

$$f(x) = \sum_{t=1}^T \hat{w}_t f_t(x).$$

We can assume that $f(x)$ is never exactly zero.

Show that

$$\varepsilon_{\text{training}} := \frac{1}{n} \sum_{i=1}^n 1_{\{F(x^{(i)}) \neq y^{(i)}\}} \leq \frac{1}{n} \sum_{i=1}^n \exp(-f(x^{(i)})y^{(i)}),$$

where $1_{\{F(x^{(i)}) \neq y^{(i)}\}}$ is 1 if $F(x^{(i)}) \neq y^{(i)}$ and 0 otherwise.

- (b) [8 points] The weight for each data point i at step $t+1$ can be defined recursively by

$$\alpha_{t+1}^{(i)} = \frac{\alpha_t^{(i)} \exp(-\hat{w}_t f_t(x^{(i)})y^{(i)})}{Z_t},$$

where Z_t is a normalizing constant ensuring the weights sum to 1

$$Z_t = \sum_{i=1}^n \alpha_t^{(i)} \exp(-\hat{w}_t f_t(x^{(i)})y^{(i)}).$$

Assume that the initial weights are uniform: $\alpha_1^{(i)} = 1/n$.

Show that

$$\frac{1}{n} \sum_{i=1}^n \exp(-f(x^{(i)})y^{(i)}) = \prod_{t=1}^T Z_t.$$

- (c) [9 points] We showed above that training error is bounded above by $\prod_{t=1}^T Z_t$. At step t the values $Z_1, Z_2, \dots, Z_{t-1}$ are already fixed therefore at step t we can choose $\hat{w}_t$ to minimize Z_t . Let

$$\varepsilon_t = \sum_{i=1}^n \alpha_t^{(i)} 1_{\{f_t(x^{(i)}) \neq y^{(i)}\}}$$

be the weighted training error for the weak classifier $f_t(x)$. Then we can re-write the formula for Z_t as

$$Z_t = (1 - \varepsilon_t) \exp(-\hat{w}_t) + \varepsilon_t \exp(\hat{w}_t).$$

- (i) [3 points] First find the value of $\hat{w}_t$ that minimizes Z_t . Then show that the corresponding optimal value is

$$Z_t^{\text{opt}} = 2\sqrt{\varepsilon_t(1 - \varepsilon_t)}.$$

- (ii) [3 points] Assume we choose Z_t as Z_t^{opt} in part (i). Then re-write $\varepsilon_t = 1/2 - \gamma_t$, where $\gamma_t > 0$ implies better than random and $\gamma_t < 0$ implies worse than random. Then show that

$$Z_t \leq \exp(-2\gamma_t^2).$$

(You may want to use the fact that $\log(1 - x) \leq -x$ for $0 \leq x < 1$.)

- (iii) [3 points] Finally, show that if each classifier is better than random, i.e., $\gamma_t > \gamma$ for all t and $\gamma > 0$, then

$$\varepsilon_{\text{training}} \leq \exp(-2T\gamma^2),$$

which shows that the training error can be made arbitrarily small with enough steps.